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A-15
B-25.5
40.5

THE COPPERBELT UNIVERSITY
SCHOOL OF MEDICINE
End of term test: MAY 2025
MBS 240
Term 1- Laboratory Sciences

Name: _____

Student ID Number: _____

Program: MBCHB

Time: 1 hr and 15 minutes.

Instructions:

Section A: Multiple choice Questions (20 marks).

Choose one correct answer and circle the letter in front of it. e.g. (C)
There are 20 questions
Answer all questions

Section B: Short answer questions (30 marks).

Write short notes on 3 out of the 5 questions.
10 Marks are allocated to each question.
Write your Student ID number on top of each page of the answer book.

Section A. Multiple choice questions.
Single answer questions

1. In Laboratory Medicine what information is obtained from the question "What is the patient's prognosis"?

- a) Exclude a diagnosis
- ☒ b) A diagnosis is being sought
- c) Rule out a condition
- d) Assess the outlook of the patient
- e) Concerns about the patient management.

2. Which one of the following is not an essential feature of a medical laboratory Standard Operating Procedure Manual?

- a) It is strictly adhered to by all in the laboratory
- b) It is periodically reviewed and updated
- ☒ c) It has laboratory Safety Instructions
- ☒ d) It is accurate
- e) It is readily available

3. Which one of the following is NOT correct about requesting a laboratory test in a resting state?

- ☒ a) It is useful in detecting minor abnormalities of a system
- b) It is useful in investigating secretory function of an organ
- c) It is useful in investigating excretory function of an organ
- d) It is useful in detecting major abnormalities of a system
- e) None of the above

4. Which is the main body compartment in the analysis of chemical changes responsible for a Biochemical disorder?

- a) Intracellular fluids
- b) Lymph
- c) Intestinal content
- ☒ d) Plasma
- e) None of the above

5. One of the following falls within the normal proportion of HbA2 in adult humans:

- a) 10%
- b) 5%
- ☒ c) 2%
- d) 1%
- e) None of the above

6. Which is the commonly used anticoagulant in haematology?

- a) Warfarin
- ☒ b) Ethylene Diamine Tetra Acetic Acid
- c) Magnesium citrate
- d) Sodium chloride
- e) Potassium oxalate

7. Acarology is the study of:

- a) Insects
- ☒ b) Single celled parasites
- c) Many celled parasites
- d) Ticks
- e) Vectors

8. A host that is not necessary for the completion of a parasite's life cycle is called:

- a) Definitive host
- b) Paratenic host
- ☒ c) Intermediate host
- d) Vector
- e) Reservoir host

9. The following is TRUE about Bacteria, EXCEPT,

- a) Gram positive bacteria have a thick peptidoglycan layer
- b) They can be rod shaped
- ☒ c) Gram negative bacteria have teichoic acid
- d) They reproduce asexually by binary fission
- e) Some are acid fast

10. Which of the following bacteria is an acid-fast bacterium?

- a) *Clostridium*
- b) *E. coli*
- c) *Treponema pallidum*
- ☒ d) *Mycobacterium tuberculosis*
- e) *Vibrio cholerae*

11. The following diseases are caused by bacteria, EXCEPT

- ☒ a) Ebola
- b) Tuberculosis
- c) Typhoid fever
- d) Cholera
- e) Plague

12. The following is TRUE about bacterial sexual reproduction, EXCEPT,

- a) In Transformation, DNA is acquired directly from the environment
- b) In Transduction, a Bacteriophage transfers the genes between bacteria
- c) In Conjugation, DNA crosses a sex pilus from donor to recipient
- ☒ d) In Translation, RNA is acquired directly from the environment
- e) In Conjugation, there is a cell to cell contact of bacteria

13. The following statements on prokaryotic cells are correct except

- ☒ a) Cells have a membrane around the nucleus.
- b) Most cells are unicellular.
- c) Bacteria are a common example of prokaryotic cells.
- d) The plasma membrane typically has no carbohydrates.
- e) The chromosome is singular and lacks histones.

14. Eukaryotic cells

- a) Have a simple flagella with two protein building blocks.
- b) Reproduce sexually through mitosis.
- ☒ c) Have a chemically simple cell wall if present.
- d) Are typically 2 μm in diameter.
- e) Do not have membrane bound organelles

15. With regards to microscopic methods

- a) Fluorescent microscopy has a higher resolution than an electron microscopy.
- b) All microscopes require staining of the specimen before visualisation.
- c) Brightfield microscopy creates a 3D image.
- d) Darkfield microscopy causes the specimen to appear dark against an illuminated background.
- ☒ e) The phase contrast microscope relies on the principle of light being dispersed according to specimen density.

16. During slide preparation for microscopic examinations

- a) Fixation is the treatment of cells with a mild surfactant which dissolves the cell membrane.
- ☒ b) Auramine-rhodamine is fluorescent dye that stains acid fast organisms.
- c) The gram stain is used to detect blood parasites.
- d) Potassium hydroxide is used as a contrast stain.
- e) India ink is used to differentiate gram positive bacteria from gram negative bacteria.

17. Which one of the following is the smallest virus?

- a) Poxvirus
- ☒ b) Parvovirus
- c) Influenza virus
- d) Rabies virus

18. Which one of the following viruses produces Negri bodies?

- a) Adenovirus
- b) Influenza virus
- ☒ c) Rabies virus
- d) Herpes simplex virus

19. The Ebolavirus is

- a) Bullet shape
- b) Rod shape
- c) Brick shape
- ☒ d) Filamentous shape

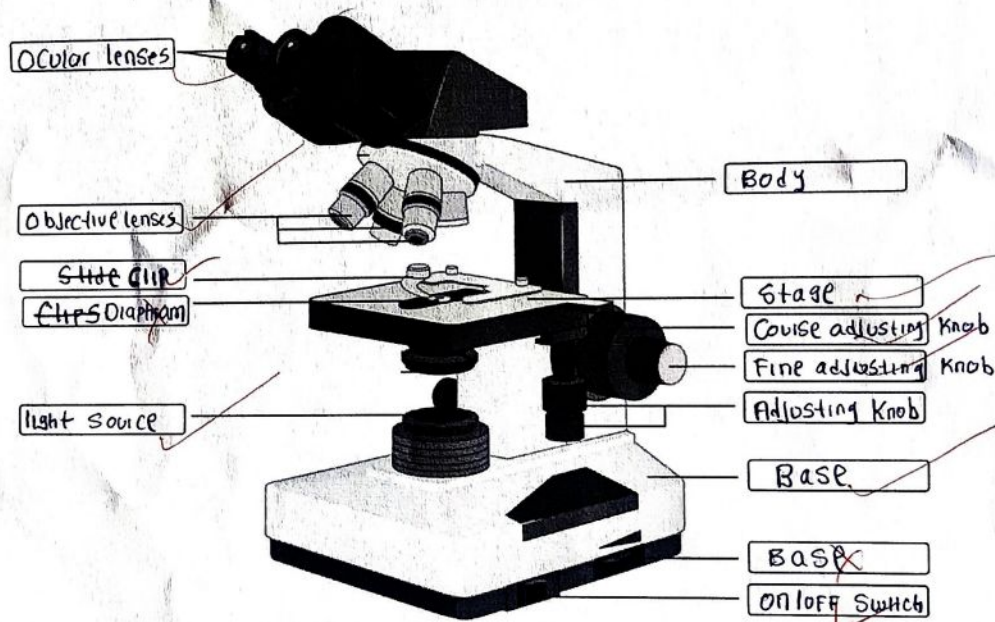
20. Which of the following statements most accurately distinguishes biosimilars from small molecule drugs?

- ☒ a) Biosimilars are chemically synthesized, while small molecule drugs are derived from living organisms.
- b) Small molecule drugs have high molecular weight, whereas biosimilars have low molecular weight.
- c) Biosimilars are derived from living organisms and have a more complex structure than small molecule drugs, which are chemically synthesized and have a well-defined structure.
- d) Both are derived from living organisms, but biosimilars are less complex.
- e) Small molecule drugs are exclusively used for infectious diseases, while biosimilars are for chronic conditions.

Section B. Short answer questions.

Choose 3 out of the 5 questions.

1. Which questions would you want answered when requesting laboratory investigations for a Biochemical disorder in a patient? For each question describe the information that is provided.
2. Define the term "Reference range" in clinical chemistry. Which factors affect this range?
3. With the aid of a diagram, name the parts of a bacterial cell and the specific function of each part in the cell
4. Label the parts of the microscope and describe the function of each part.



5. By drawing a graph explain the sequential differences in the magnitude and characteristics of the innate versus adaptive immune response during a primary infection compared to a secondary infection with the same antigen.

END OF TEST

Section B

Question 2

Reference Range

Introduction

The Reference range also called the normal range is the established set of values that describes the minimal and maximal values of body fluids in a healthy person.

It is applied to all body fluids in the body.

Factors Affecting RR

These factors can alter the reference by decreasing, maintaining or increasing it. They include;

1 Methodology

This focus on the analytical methods used when determining the RR. It says that values may not be entirely accurate, precise or specific.

Therefore in the interpretation of results, variability should be considered.

2 Physiological

These are non-pathological investigations, they include

- (a) The menstrual cycle
- (b) Time of the day
- (c) General diet

3 Race

This part can be used to describe the nutritional and environmental aspects to explain variations in RR.

Genetics can also be used to highlight the Reference Range.

4 AGE and SEX

Body fluids differ according to the age of the person e.g. in infants, adults or the old aged, It's also established that these body fluids vary according to sex i.e. male body fluids are different from females.

Conclusion

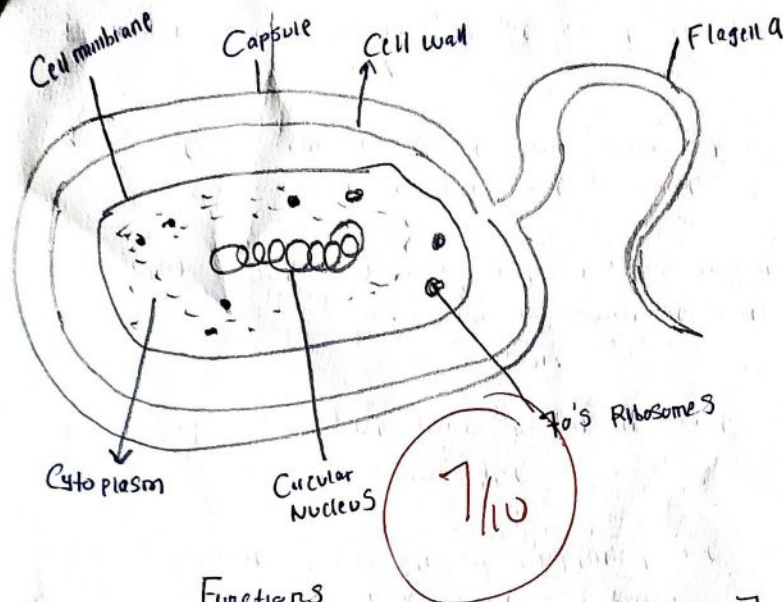
The Reference range is very important in Clinical Chemistry because it can be used to provide a differential diagnosis by measuring an individual's body fluids then comparing them to it.

FUNCTIONS OF THE PARTS OF MICROSCOPE

1. Ocular lens - magnifies the image of the objective lens usually 10X
2. Slide - This is where the specimen is mounted for examination under the microscope's stage
3. light source - provides rays of light which can be focused and directed to illuminate the specimen under observation.
4. Body - holds the lens system in place and is used to lift the microscope
5. Objective lenses - used to magnify the specimen. we have the 10X, 40X and 100X
6. Clips - hold the specimen's slide in place on the stage
7. Course adjusting knob - makes big movements to the stage to align the specimen under observation.
8. Fine adjusting knob - makes fine and small movements to the stage to bring the specimen into focus (Fine Focusing)
9. Base - supports the entire frame of the microscope. also used for lifting
10. Stage - This is where the slide is mounted
11. Diaphragm - further amplifies and modifies focused light from the condenser onto the specimen.

8.5

Question 3



Functions

[genome]

Nucleus - holds the cell's genetic information and controls activities of cell.

Flagella - made of flagellin protein is used for movement, propels the bacteria.

Capsule - forms the glycocalyx [glycoproteins / glycolipids] and is used for attachment to surfaces i.e. hosts. It protects the bacteria from damage. It has pili onto its surface which is used to attach to the host.

Cell membrane - for exchange of material i.e. selects what goes in and out i.e. minerals, waste, nutrients etc.

Cytoplasm - has no cytoskeleton and no cytoplasmic streaming. It's where activities i.e. metabolic reactions take place and contains dissolved membrane-unbound organelles i.e. ribosomes.

70's Ribosomes - used in protein synthesis.

Cell wall - made of peptidoglycan, which is used to differentiate Gram (+ve) and Gram (-ve) bacteria.

**MEN IN
FLAMES**



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